BATPING Progress Report

For the week of July 13–19, 2026

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Abstract

This note summarises BATPING bench-prototype progress during the week of July 13–19, 2026. Three activities are reported: (1) piezo-array transfer function re-characterisation for both the UltraMic 384k and ECM999 microphones; (2) new chirped-waveform recording sessions with both microphones and the diagnosis and correction of three matched-filter pipeline failure modes that produced jump discontinuities in the differential time-of-flight (TOF) time series; and (3) a first attempt at a controlled temperature-change experiment using a resistive dummy load on the optical rail, along with the issues identified and the planned course of action for corrected heat tests during the upcoming week.

1 Piezo Array Transfer Function Measurements

I re-recorded transfer functions for both the piezo-array + UltraMic 384k and piezo-array + ECM999 systems. Each frequency step used an 8 ms sine wave[1], with 20 ms settle time between steps and 50 Hz spacing between frequencies (resulting in 901 frequencies from 20 kHz to 65 kHz measured). This yielded per-step SNR of 30–60 dB across the piezo passband. Figure 1 compares the two resulting system transfer functions.

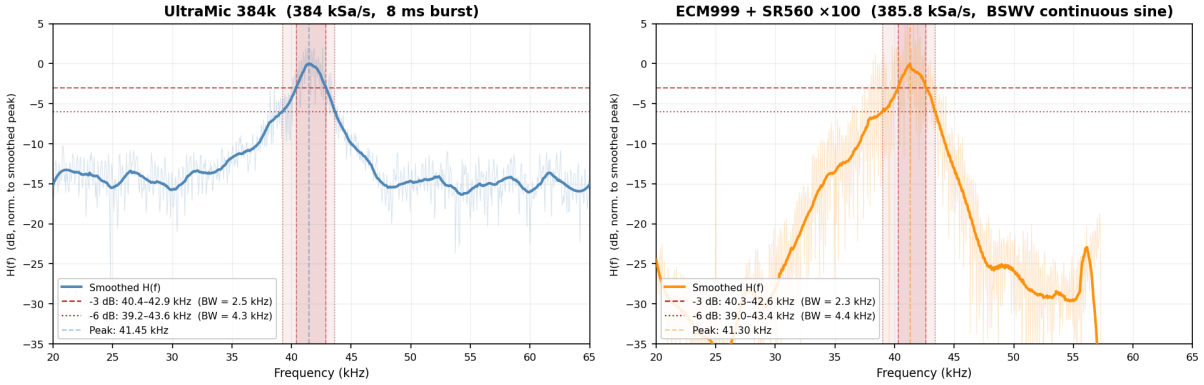


Figure 1: Piezo-array system transfer function $H(f)$, normalised to the smoothed peak, measured via stepped-sine with the UltraMic 384k (*left panel*) and the ECM999 + SR560 ×100 preamplifier connected to a PicoScope 5444D (*right panel*). Thin traces show the raw FFT amplitude at each of the 900 frequency steps; bold traces show a smoothed envelope. SDG output amplitude was 1.2 V_{pp} for the Ultramic burst and 0.5 V_{pp} for the ECM continuouse sine measurements.

[1] *ecm999_stepped_sine.py* was originally written for burst mode, with `BURST_DURATION_MS=8.0`, but uploading 901 custom ARB waveforms to the SDG slowed it down relative to the faster Picoscope, leading to timeouts. I switched the script to send a continuous sine wave (single SCPI command) per frequency, but I will be able to replace this with 8ms burst using native waveforms on the SDG rather than waveform upload.

Both systems are dominated by the piezo array's acoustic resonance near 40 kHz. The piezo-array + ECM999 system exhibits a passband (−3 dB BW = 2.3 kHz, centered at 41.3 kHz) comparable to that attained with the UltraMic (−3 dB BW = 2.5 kHz, centered at 41.45 kHz). At −6 dB the two systems remain consistent. While the passband is relatively unaffected by the change in microphone, the further off-resonance regimes of the ECM999 transfer function on either side of the passband are significantly dampened, contrasted with the stable plateau of the Ultramic around the -15dB range. This narrowing is consistent with the ECM999's flatter intrinsic response, as the ECM999 was designed for general usage at audible frequency ranges up to 20kHz. Recording with the ECM999 preserves the piezo's sharp resonance character, but lacks the boosting of the surrounding spectral content characteristic of the Ultramic, which was designed for ultrasonic frequencies. In addition, the ECM exhibits a sharp rolloff at ≈57kHz; this is most likely the upper limit of the ECM's frequency capability.

With the new transfer functions in hand, I recomputed the pre-compensation amplitude values for all chirped waveforms, using them in the recording sessions described below in Section 2.

2 Chirped Waveform Sessions and Pipeline Improvements

Using the updated pre-compensation, I recorded two new sessions with each microphone (UltraMic 384k and ECM999). I tested burst durations of 2–4 ms, bandwidths of 6–20 kHz, and pre-emphasis exponents $\alpha = 0.2$–0.8. **Across all four sessions, we continue to achieve per-ping LS-fit measurement uncertainty $\delta(\Delta\tau)$ on the order of 2–5 ns with both microphones.**

Depsite the excellent measurement uncertainty results, several failure modes in the windowing and analysis functions of `matched_filtering_pipeline` still persist that contribute to measurement accuracy, even while measurement uncertainty remains pristine. While the Least Squares shape-fit eliminated the constant measurement jitter seen previously, many recordings display jump discontinuities in the differential-TOF time series. An example of these discontinuities can be seen in Figure 2:

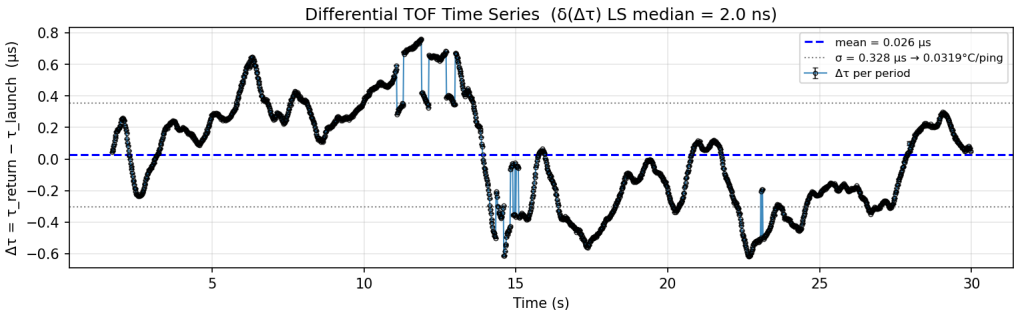


Figure 2: Differential-TOF time series $\Delta\tau(t)$ from pipeline v1 for waveform `bw_10dB_asym_2ms_comp_light` (ECM999 + SR560×100, Jul. 16 full-sweep session), showing multiple jump discontinuities scattered throughout the 30-second recording on an otherwise smooth underlying trend. **Figure 5 shows the massive improvement in data quality gained from applying the v4 pipeline to this waveform.**

The jump discontinuities arose from three distinct failure modes, illustrated in Figures 3 and 4. First, *burst-window drift*: the waveform generator and digitizer operate on independent clocks with a ≈246 ppm frequency offset [2], so over thousands of pings the acoustic burst gradually migrates within its fixed analysis window; once the burst drifts far enough to be clipped at the window boundary, the cross-correlation returns a spurious lag (visible as periods labeled "BAD" in Figure 3, where the reflected-signal trace in the analysis window is nearly zero). Second, *window overlap*: the outgoing and reflected burst analysis windows were initially placed too close together, allowing the piezo ring-down of the outgoing pulse to bleed into the reflected-pulse template. Third, *carrier-cycle slipping*: because the cross-correlation function has sidelobes spaced one carrier period (25 μs) apart, the estimator occasionally locked onto the wrong sidelobe, producing a lag error that is an integer multiple of 25 μs. Each failure mode was individually diagnosed and patched in pipeline updates v2-v4, as shown below:

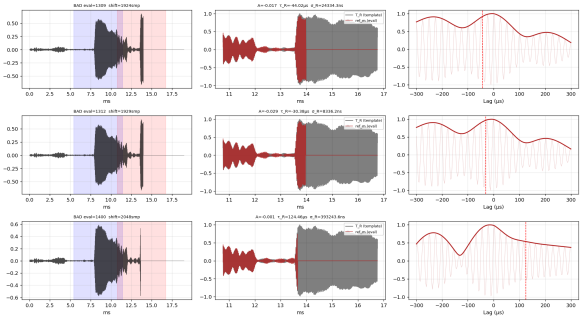


Figure 3: Per-period diagnostic for waveform `a0.8_sm_4ms` under pipeline `matched_filter_v1`, showing three representatively "BAD" evaluation periods. *Left*: drift-corrected full period with incident (blue) and reflected (red) pickup windows, displaying unwanted overlap. *Center*: co-added return template T_R (red) overlaid on the actual return signal m_{ref} (grey), displaying premature signal clipping. *Right*: Hilbert envelope of the reflected cross-correlation, displaying the LS fit latching onto noise.

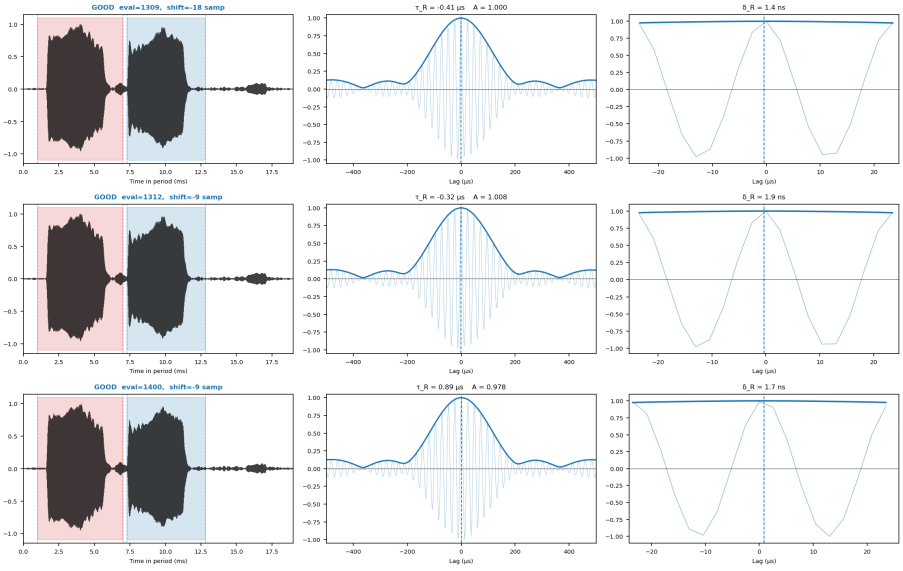


Figure 4: Same diagnostic as Figure 3 for the same three evaluation periods, but now run with the `matched_filtering_v4` pipeline. The adaptive period slicer now anchors each analysis window to the detected incident burst rather than using fixed time slots, so accumulated clock drift no longer displaces the reflected burst.

[2] At this clock offset, drift accumulation over a 30s recording totals 6.95ms. See Drift and Jitter section of the Confluence page for a full derivation of this drift behavior.

With these corrections in place, pipeline v4 eliminates almost all instances of measurement jitter across the 2 m bench path, as seen in Figure 5:

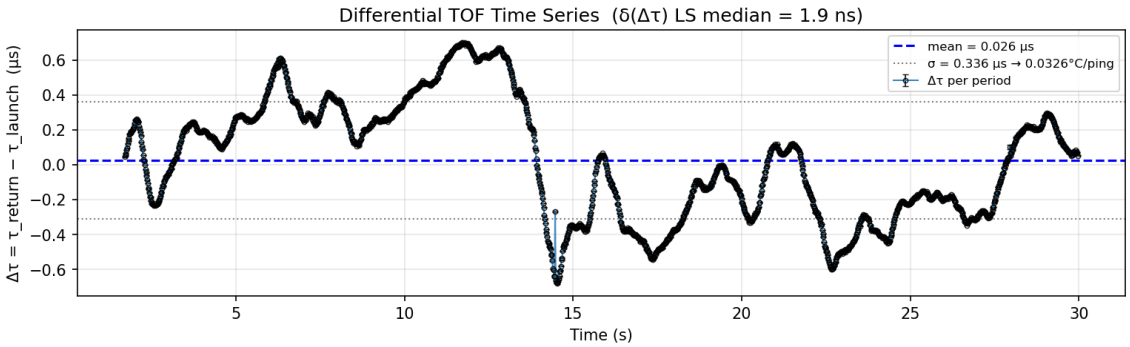


Figure 5: Differential-TOF time series from pipeline v4 for the same waveform and session as Figure 2 (`bw_10dB_asym_2ms_comp_light`, ECM999). The jump discontinuities visible in v1 are eliminated; the time series follows a smooth physical trend with the per-ping uncertainty $\delta(\Delta\tau)$ even reduced by a tenth of a nanosecond.

In addition to fixing small-scale (10^2 ns) measurement inaccuracy, the patches implemented in `matched_filter_v4` had even more drastic improvement on waveforms that had jump discontinuities on the 10 μs and 10^2 μs scale. For example, in the case of the `a0.8_md_4ms` chirped waveform, incident and reflection window overlap in the template construction section of the matched filter pipeline caused the reflection template to be contaminated by part of the incident signal. This resulted in microsecond-scale off-target CCF fitting, as shown in Figure 6: In the

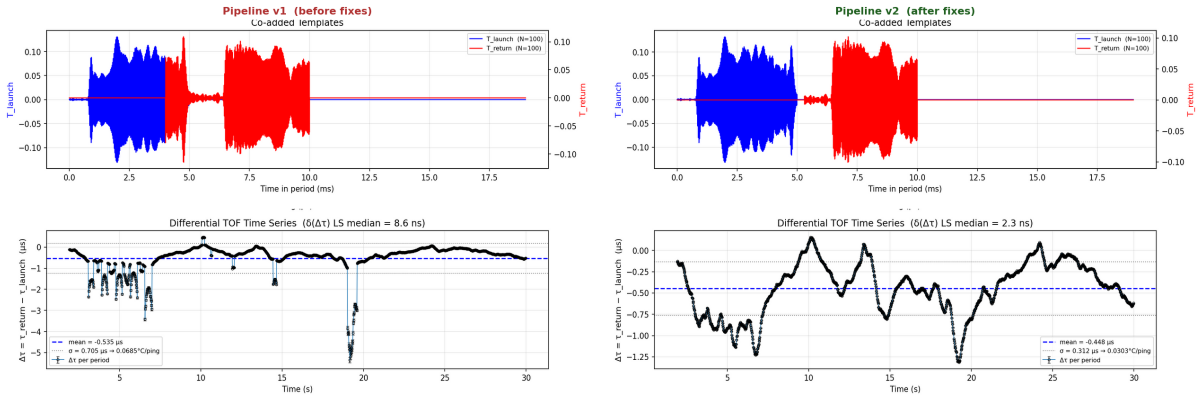


Figure 6: Differential-TOF time series from the recording of the `a0.8_md_4ms` waveform with the ECM999 microphone, showing successful resolution of window overlap failure mode; as a result, v4 pipeline cuts the v1 $\sigma_{\Delta\tau}$ value in half.

case of `a0.8_lg_2ms`, a single outlier data point inflated the $\sigma_{\Delta\tau}$ reading on the 30s timescale by a factor of three, as shown in Figure 8:

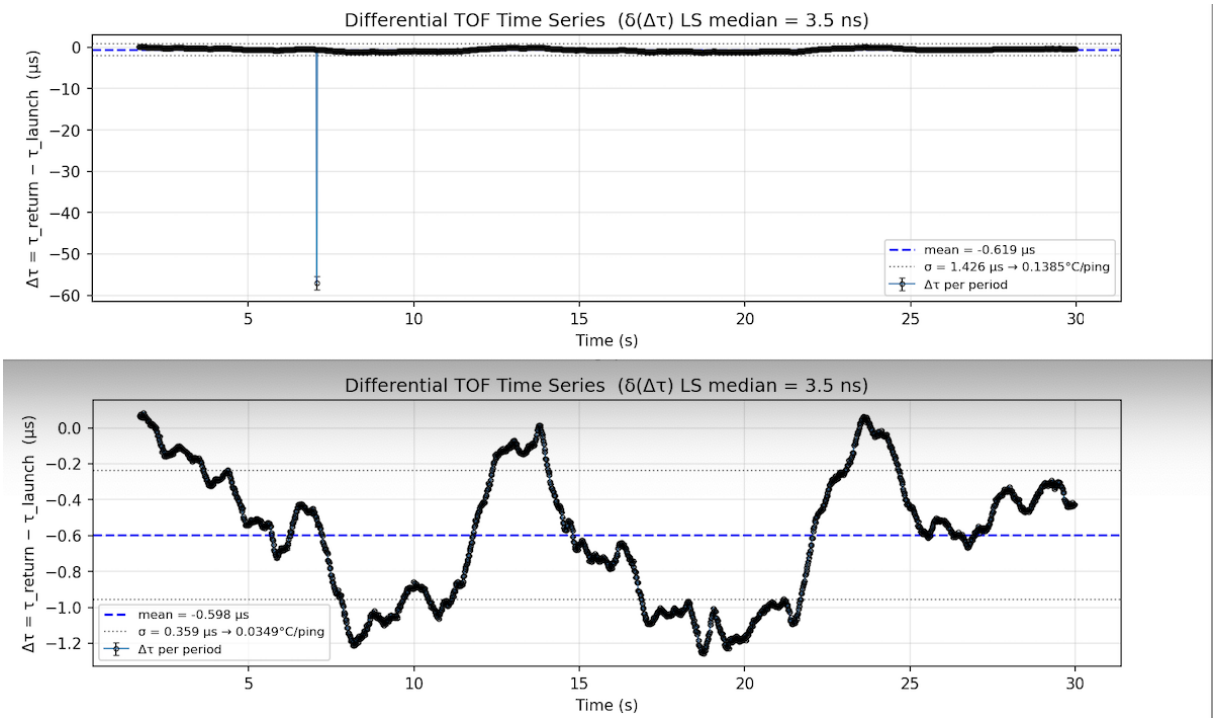


Figure 7: v1 (top) and v4 (bottom) differential-TOF time series from the recording of the `a0.8_lg_2ms` waveform with the ECM999 microphone; v1 analysis produced a large outlier data point, potentially due to a carrier cycle slip or incorrect windowing; as a result of the failure mode patches, the v4 pipeline cuts the v1 $\sigma_{\Delta\tau}$ value in third.

3 Controlled Temperature Experiment: Dummy-Load Resistor

On Thursday and Friday I ran a first controlled temperature-change test using the 25 W RF dummy-load resistor (XRDS-RF, 50 Ω, BNC connector) mounted at the midpoint of the 1 m Thorlabs optical breadboard, directly beneath the acoustic path. Figure **??** shows a picture of the physical setup. DC power was supplied by a Keithley 2231A-30-3 triple-channel supply found in the lab, and BATPING was recorded continuously using `picoscope_batping_record.py` while voltage was slowly ramped. Two power levels were explored: 10 V (≈2 W, 8% of the 25 W rating) and 30 V (18 W, 72% of rating, the Keithley's maximum output). The resistor did not heat up appreciably at 10 V after 15 minutes; it was warm to the touch at 30 V. /textbfHowever, looking back, I should have refrained from powering the resistor until it felt warm to the touch, because such a temperature fluctuation (on the scale of many degrees Celsius) is much more extreme than anything we are interested in measuring. As a result, both recordings showed nanosecond-scale uncertainty during quiet stretches, but with frequent large jump discontinuities that made the data unusable.

I think that *acoustic scintillation* from over-driving the dummy-load resistor may have been the first-order issue that caused the degraded data quality. When driven with 30 V, the resistor dissipates 18 W, which is roughly 72% of its power rating . This could have caused a stronger than anticipated convective plume to form above the rail, which would have scattered the acoustic

wavefront, causing the received signal amplitude to fluctuate by more than an order of magnitude between 30-second analysis windows. Windows with near-zero amplitude yield meaningless lag estimates regardless of pipeline quality.

A secondary issue with this data collection was that I used the *incorrect PicoScope sample rate*. The TIMEBASE register had been set using the formula for 8-bit ADC resolution, whereas the unit was operating in 12-bit mode; this resulted in an actual sample rate of 251 kSa/s (6.3 samples per 40 kHz carrier cycle) rather than the intended 1 MSa/s (25.6 samples/cycle).

Both issues have been corrected for measurement during this coming week. The recording script now uses the proper 12-bit TIMEBASE value (verified by direct SDK query to the connected scope), and the next heat test will use a much lower power level of 5–10 V (0.5–2 W). At these levels the thermal gradient should develop slowly and without a disruptive convective plume; at 1 W, an air-column temperature rise of ∼0.1–0.5 °C over 5–10 minutes is expected — well within BATPING's ∼0.05 °C per-ping sensitivity. Additionally, I will position a long strip of acoustic foam *above* the region of the acoustic path that is affected most by the dummy-load resistor. This will hopefully mitigate the natural tendency of hot air to rise, which would take it up and away from the region of space crossed by the signal. I may even attempt to form a full enclosure of the path using the foam, leaving enough room on each side to avoid extreme multipath confusion. I plan to re-record in this new setup, at lower power levels, and with proper sampling rate during the week of July 20-24 and will send an update in the coming days.

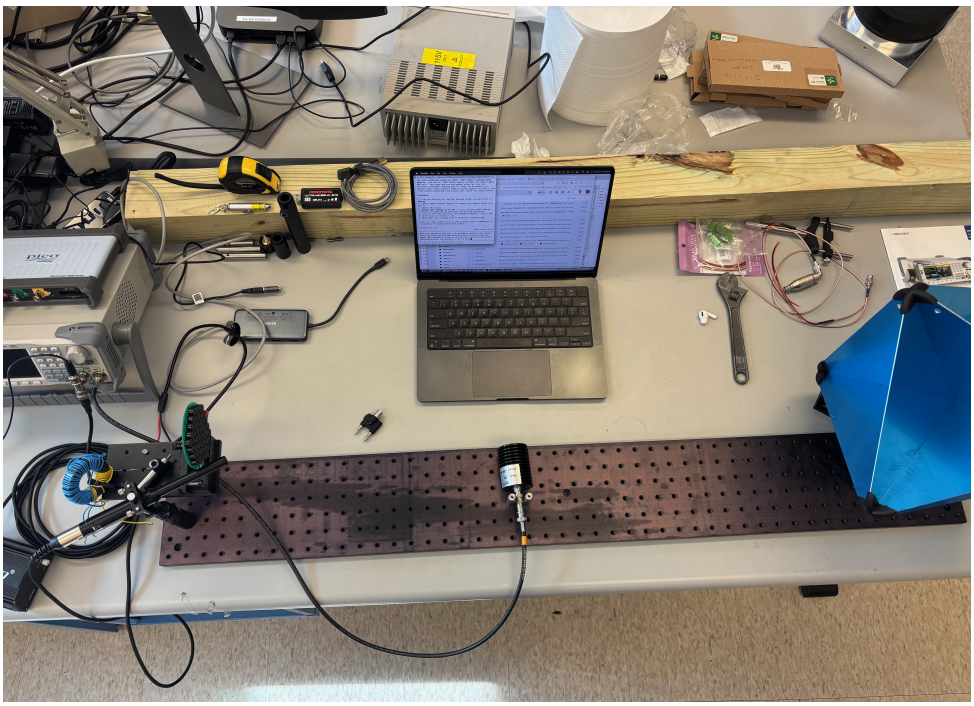


Figure 8: Picture of physical setup, showing dummy-load resistor mounted on the optical rail by two posts; the resistor is positioned ≈10cm below the acoustic path, with half of its fins facing radially outward towards the acoustic path (and half facing into the rail).